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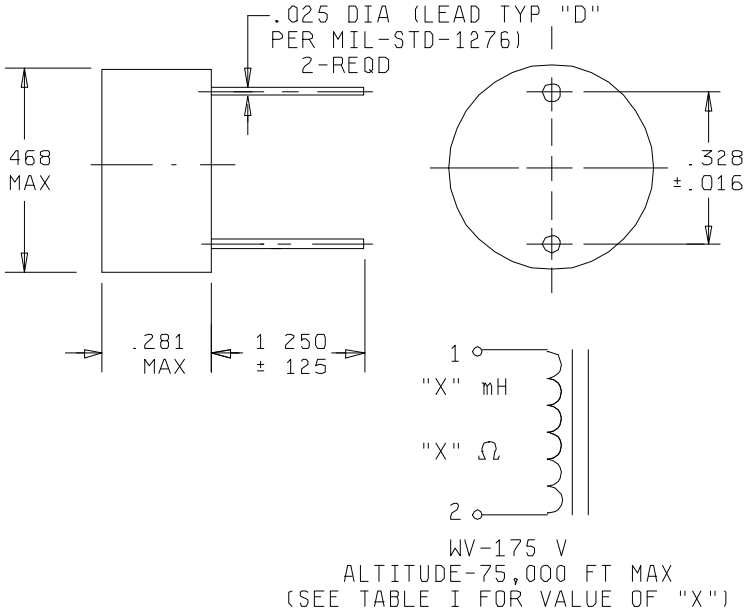
MIL-PRF-27/241A
8 April 1980
SUPERSEDING
MIL-T-27/241
11 September 1978

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE)
INDUCTORS, AUDIO FREQUENCY, HIGH Q, TF5R20ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the inductors described herein shall consist of this document and the latest issue of Specification MIL-T-27.



CIRCUIT DIAGRAM AND MARKING

INCHES	MM
.016	0.41
.025	0.64
.125	3.18
.281	7.14
.328	8.33
.468	11.89
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.

(A) Denotes changes

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings See table I

TABLE I Electrical ratings 1/

Dash No	Inductance 2/ mH (1-2) ±2%	DC 3/ current mA (1-2) max	DC resistance ohms (max) (1-2)	Quality factor (min)	Temperature stability (-55° to +105°C)	Working voltage (peak) volts	Voltage (1-2)
01	0.6	90	1.9	70 at 100 kHz	±1%	175	0.1 volt at 1 kHz
02	1 5	57	4.9	70 at 100 kHz	±1%	175	0 1 volt at 1 kHz
03	2 5	44	8 2	70 at 100 kHz	±1%	175	0.1 volt at 1 kHz
04	6 0	28	19.0	70 at 100 kHz	±1%	175	0 1 volt at 1 kHz
05	10 0	22	32 0	65 at 50 kHz	±1%	175	0 1 volt at 1 kHz
06	15.0	18	49.0	60 at 50 kHz	±1%	175	0 1 volt at 1 kHz
07	25 0	14	82.0	55 at 50 kHz	±1%	175	0 1 volt at 1 kHz
08	40.0	11	132.0	48 at 50 kHz	±1%	175	0.1 volt at 1 kHz

1/ Qualification testing and approval to M27/241-8 shall be sufficient to grant qualification approval to M27/241-1 through M27/241-7

2/ The inductance is measured with 0 Adc applied to (1-2) and at the specified voltage across (1-2)

3/ The amount of DC current that will reduce the inductance a maximum of 7%

Design and construction
Dimensions and configuration See figure 1.
Duty cycle. Continuous.
Case Diallyl phthalate
(A) Terminals 0 025 tin-lead plated type D4 in accordance with MIL-STD-1276
Weight. 0.08 ounce maximum
Altitude 75,000 feet maximum
Operating temperature range -55° to +105°C

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage:
At sea level: 500 volts rms.
At reduced barometric pressure: 300 volts rms

Vibration (high frequency) MIL-STD-202, method 204

Marking location. See figure 1

Part number M27/241- (dash number from table I)

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Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, WC
Navy - MC, AS
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0569-15)

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TF5R20ZZ

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